WORLD STATISTICS CONGRESS 2025 THE HAGUE

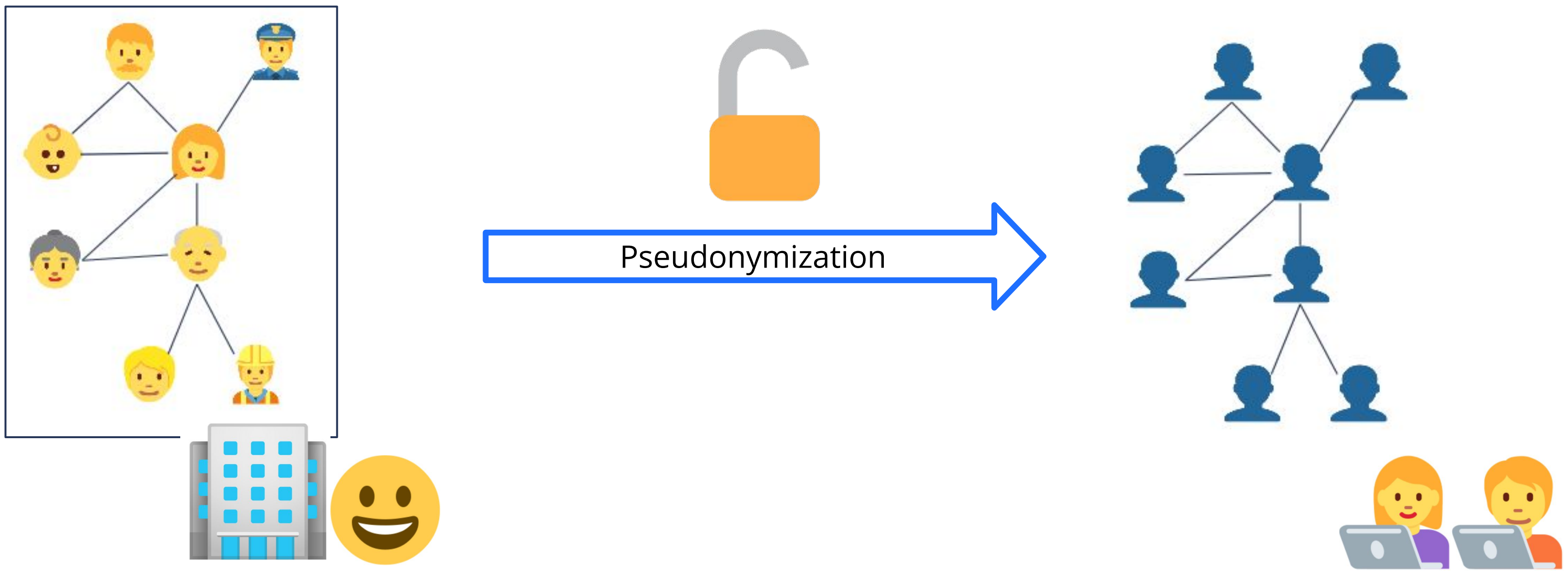
International Statistical Institute

Network Science for Official Statistics

Anonymity and Disclosure Control for Network Data

Rachel de Jong, Mark van der Loo, Frank Takes
Leiden university, CBS
October 5th 2025

Safely publishing networks?



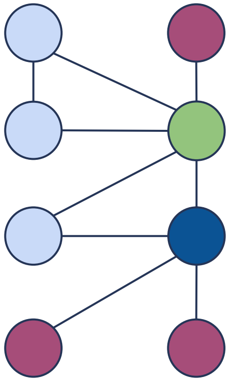
Towards Anonymity in Network Data

Microdata

ID	Attribute
1	
2	
3	
4	

ID	Attribute
5	
6	
7	
8	

Network data

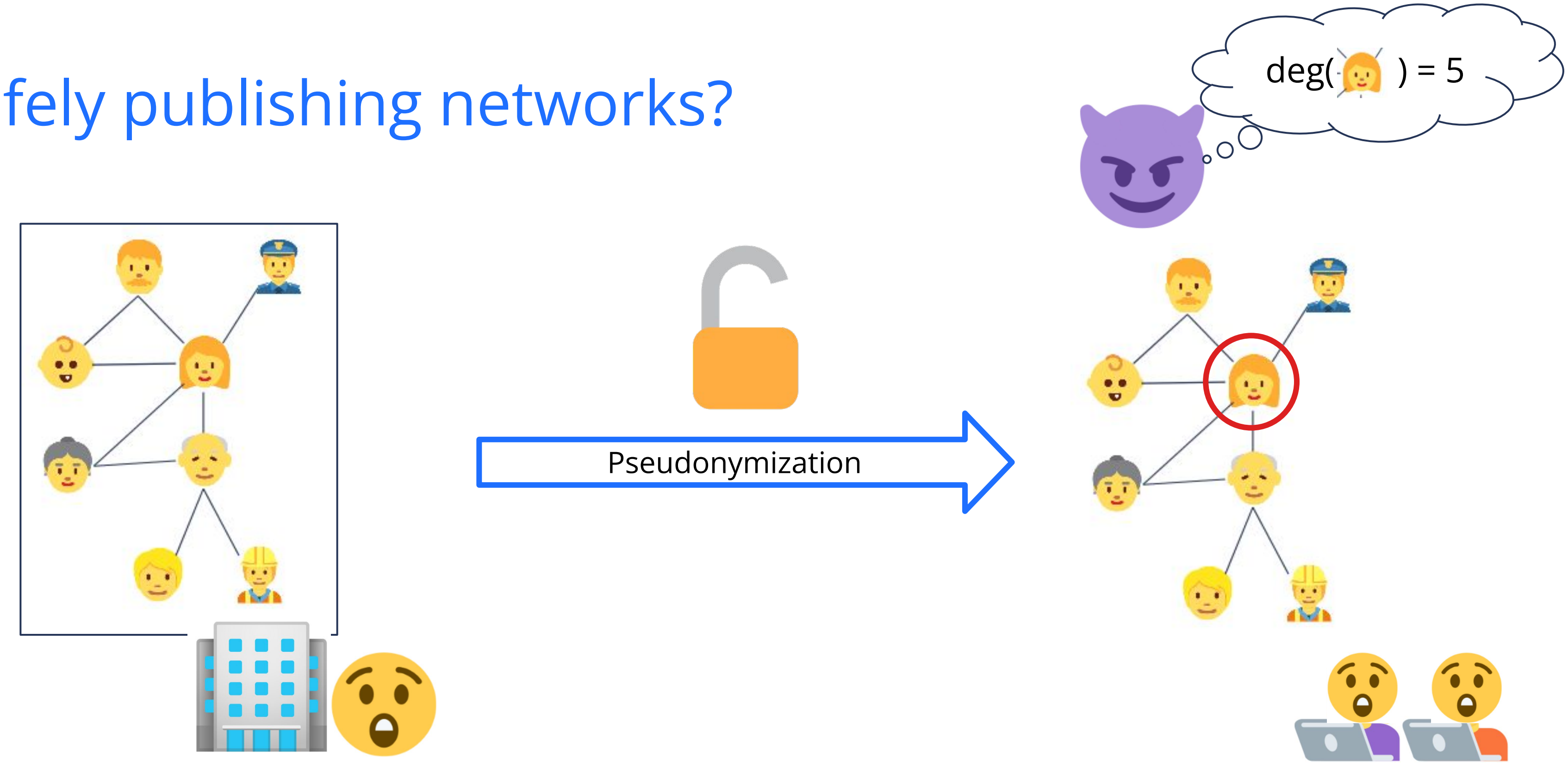


Network consists of

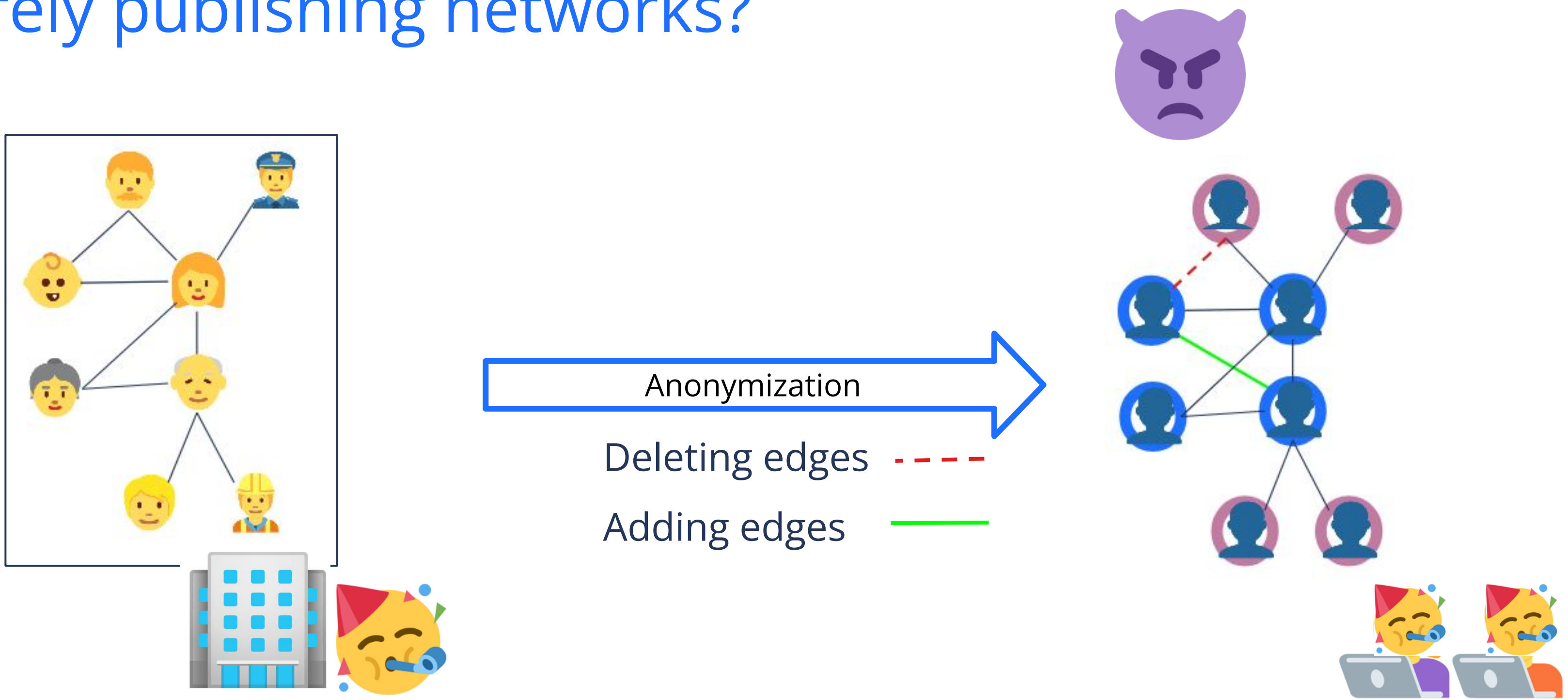
- Nodes O
  - Entities: people, companies, cities, …
- Edges —
  - **Connections**: social ties, trade, roads, …

Identification risk

Safely publishing networks?



Safely publishing networks?



Q1: How to **measure** anonymity?

- *k*-Anonymity
- Measure = attacker scenario
- Uncertain information?

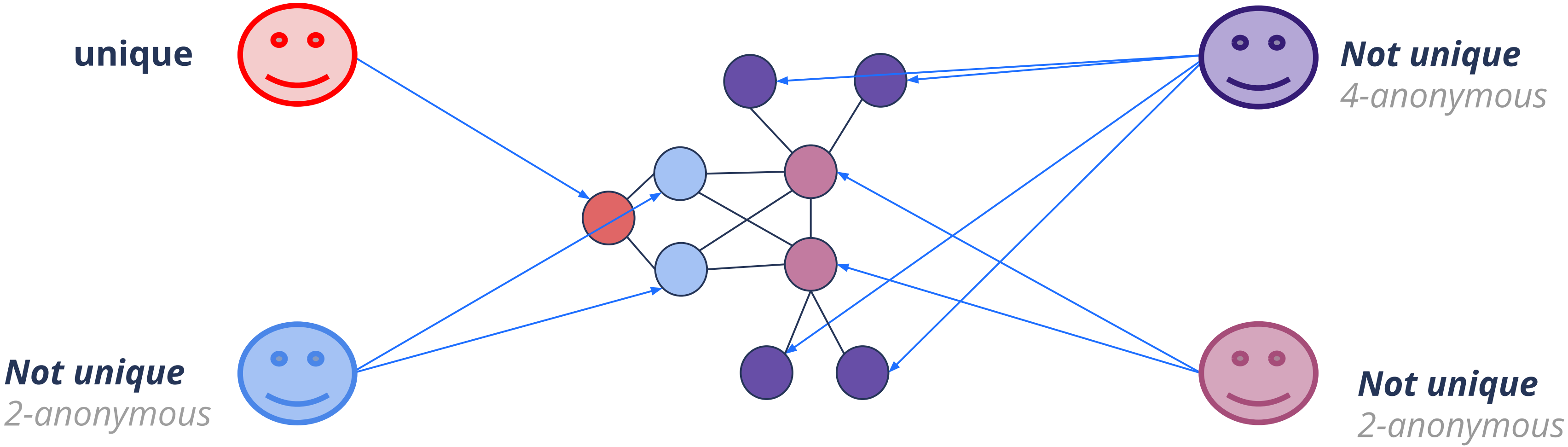
Q2: How to **anonymize** a network?

k-Anonymity

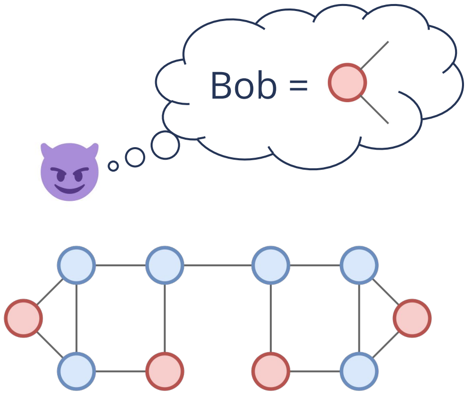
How many candidates are there for each individual?

anonymity = uniqueness: fraction of unique nodes

For now: **Not unique** → **Anonymous**

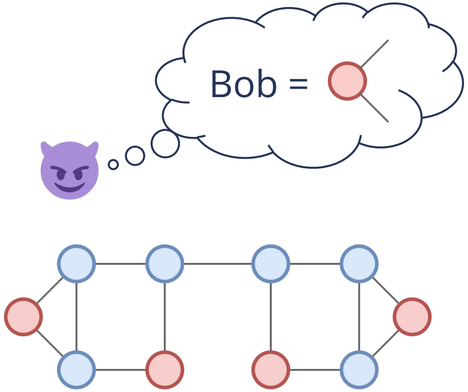


Measure = attacker scenario

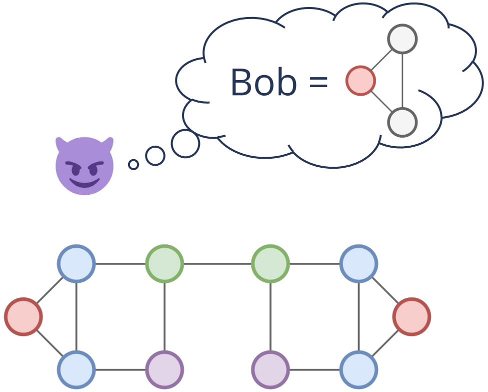


Degree

Measure = attacker scenario

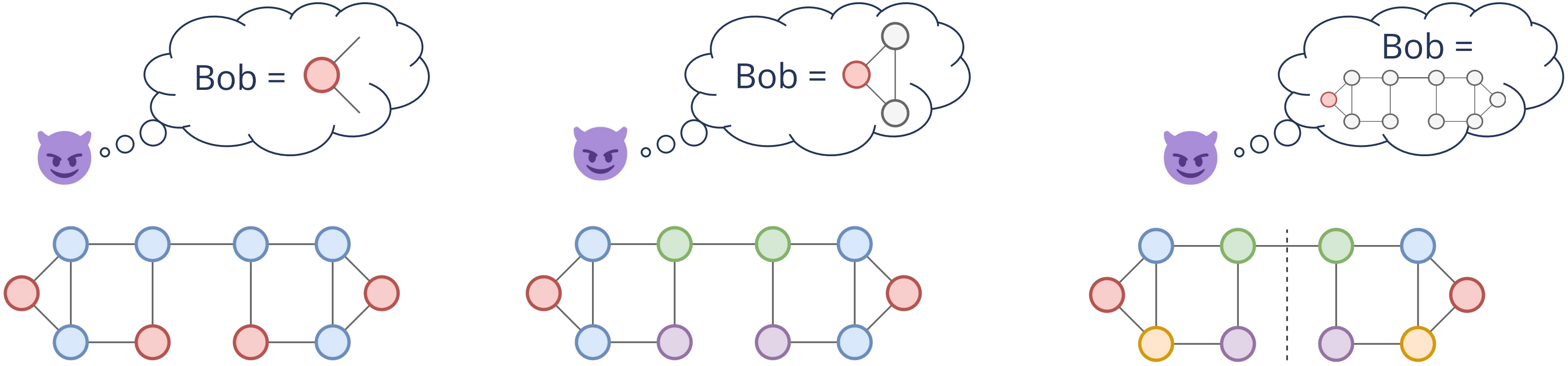


Degree



1-Neighborhood

Measure = attacker scenario

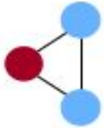


Degree

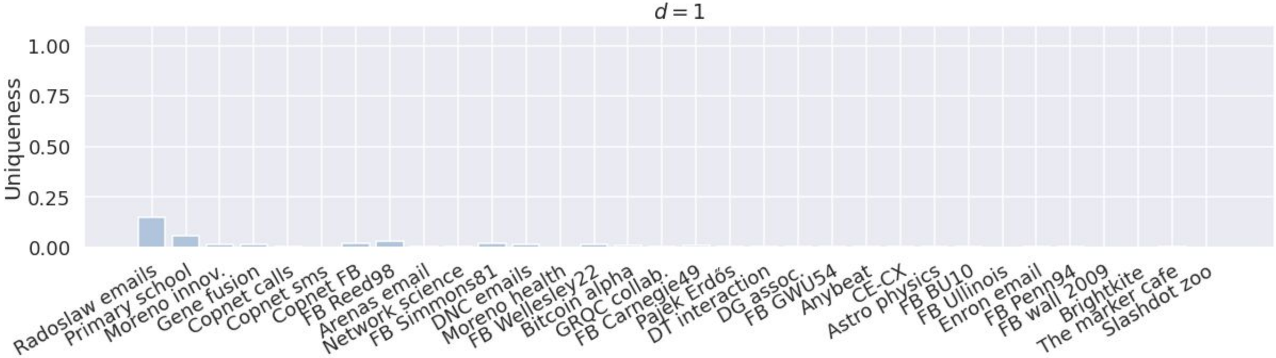
1-Neighborhood

Automorphism

Measure = attacker scenario

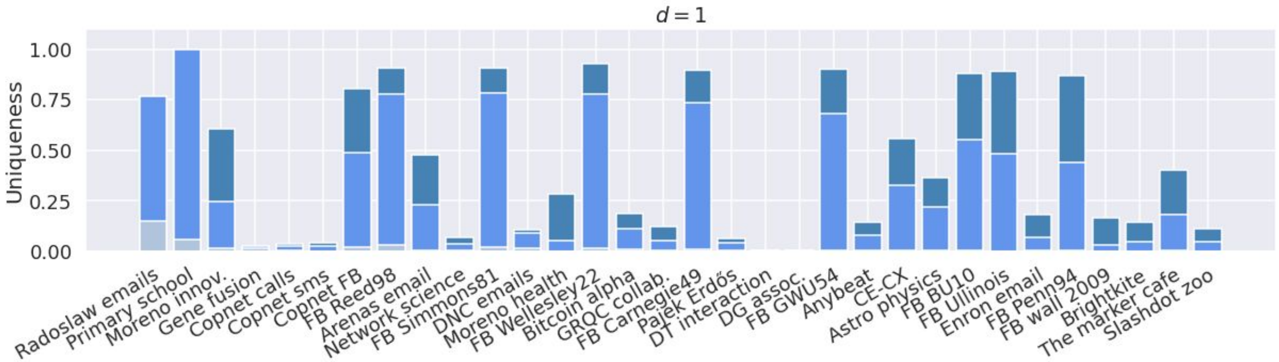


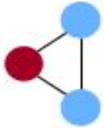
Degree 2



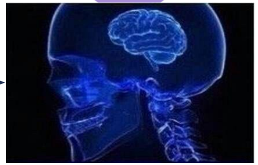
|V|: ~100 - 3M |E|: ~100 - 18M

Measure = attacker scenario



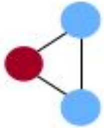


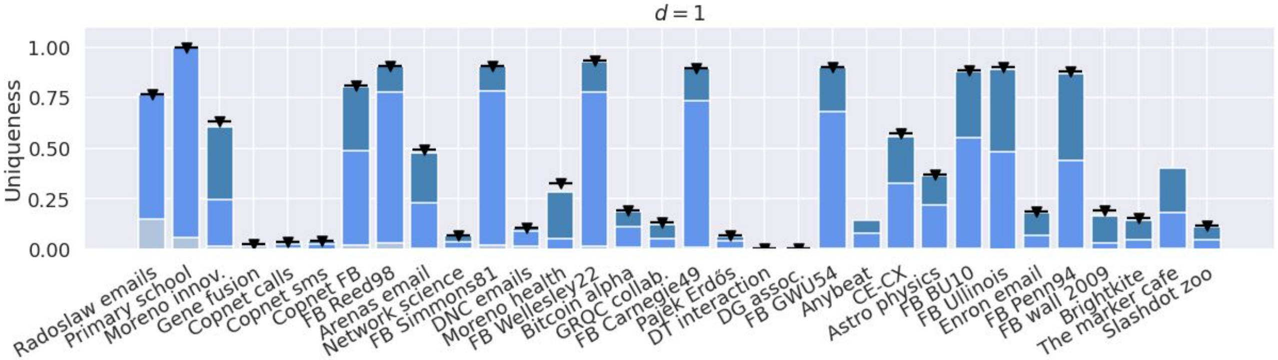
Degree	2
(|V|, |E|)	(3, 3)
Degree distribution	{2, 2, 2}



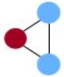
|V|: ~100 - 3M |E|: ~100 - 18M

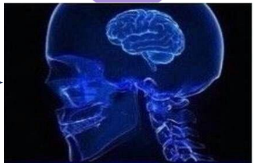
Measure = attacker scenario



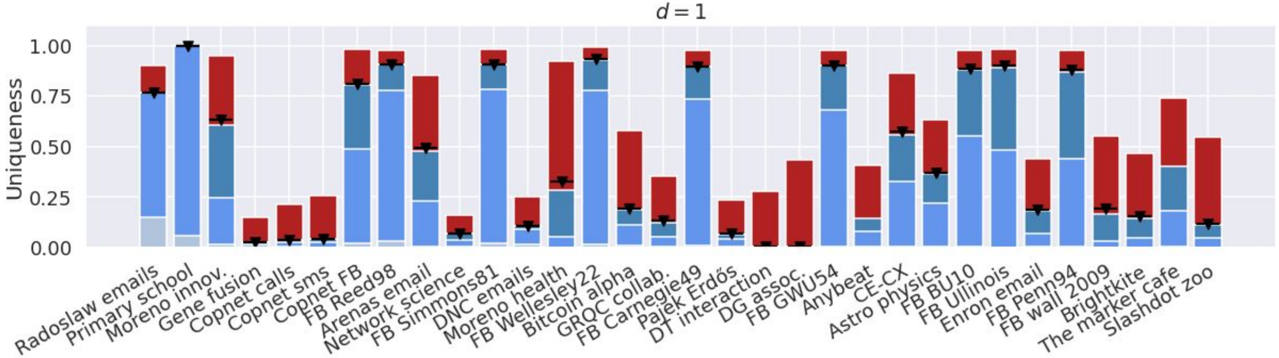


|V|: ~100 - 3M |E|: ~100 - 18M

Degree	2
(|V|, |E|)	(3, 3)
Degree distribution	{2, 2, 2}
Isomorphism	

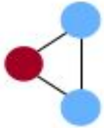


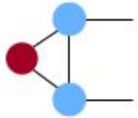
Measure = attacker scenario

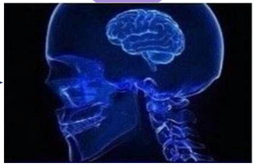


|V|: ~100 - 3M |E|: ~100 - 18M

Imprecise information but looking further
is more effective!





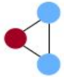


Degree 2

(|V|, |E|) (3, 3)

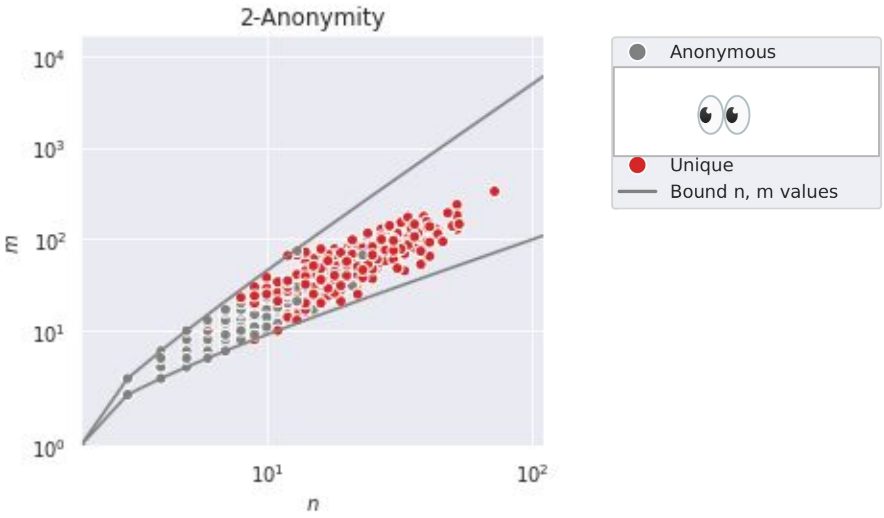
Degree distribution {2, 2, 2}

Isomorphism



Degrees of neighbors {2, 3, 3}

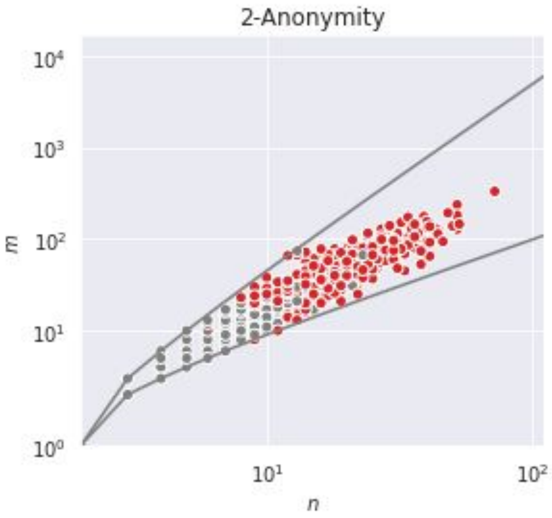
Fuzzy-k-anonymity



Arenas email network

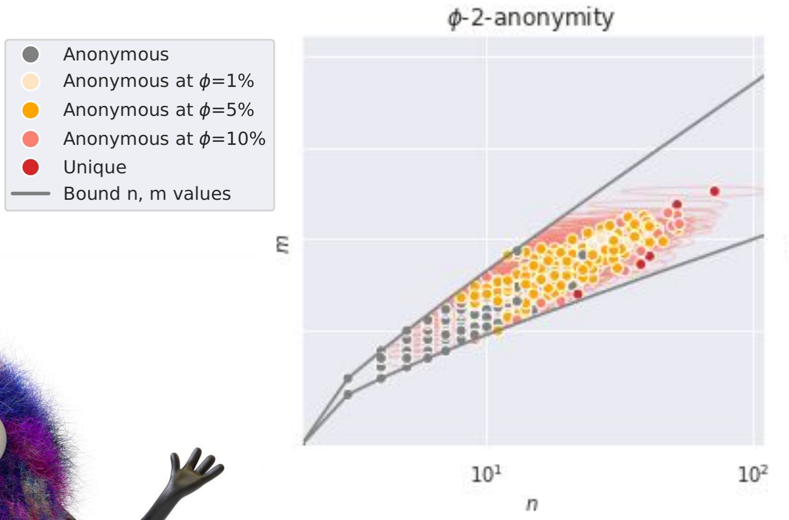
What if the attacker is not so sure?

Fuzzy-k-anonymity



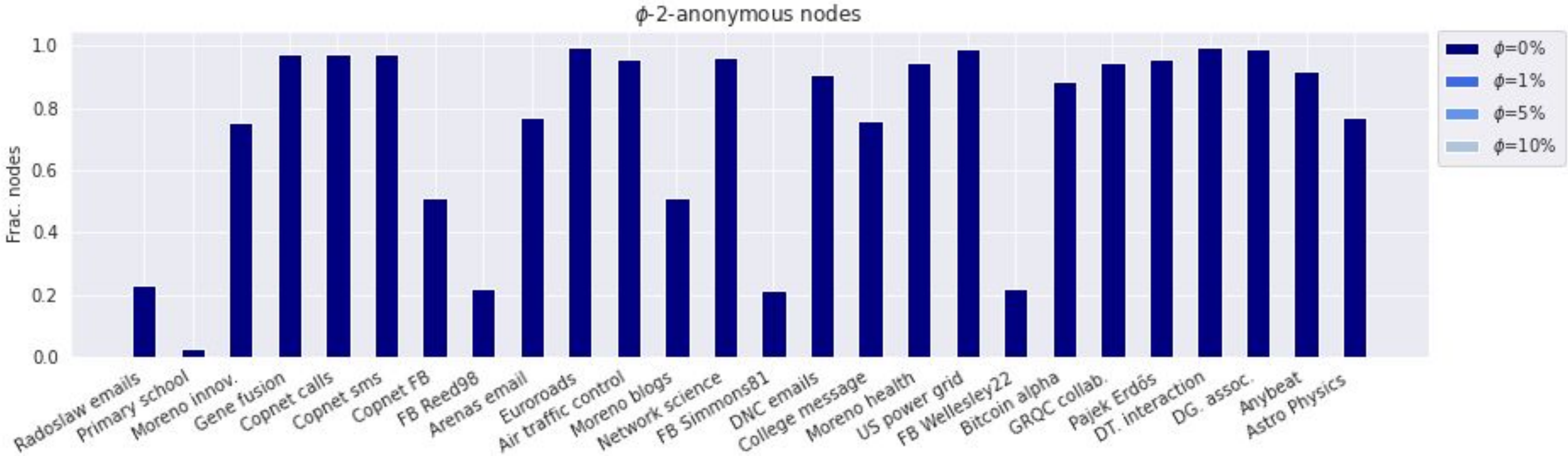
ϕ-similar if:

$$\Delta_n + \Delta_m \leq \phi(n+m)$$



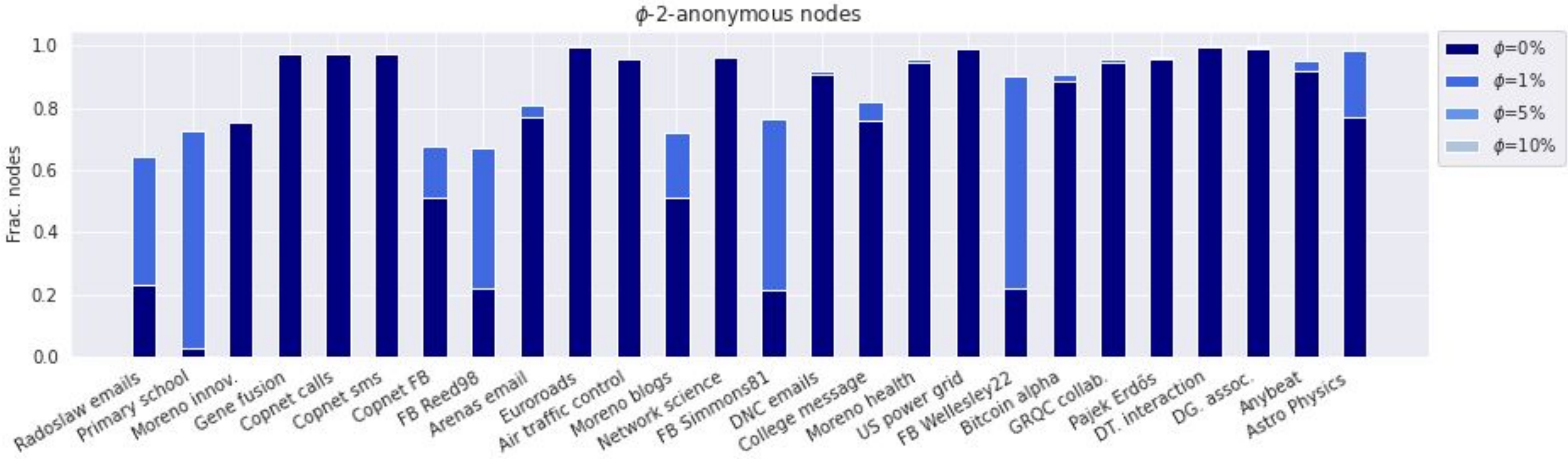
Arenas email network

Fuzzy k-anonymity in networks



No uncertainty: **>2%** anonymous for all networks

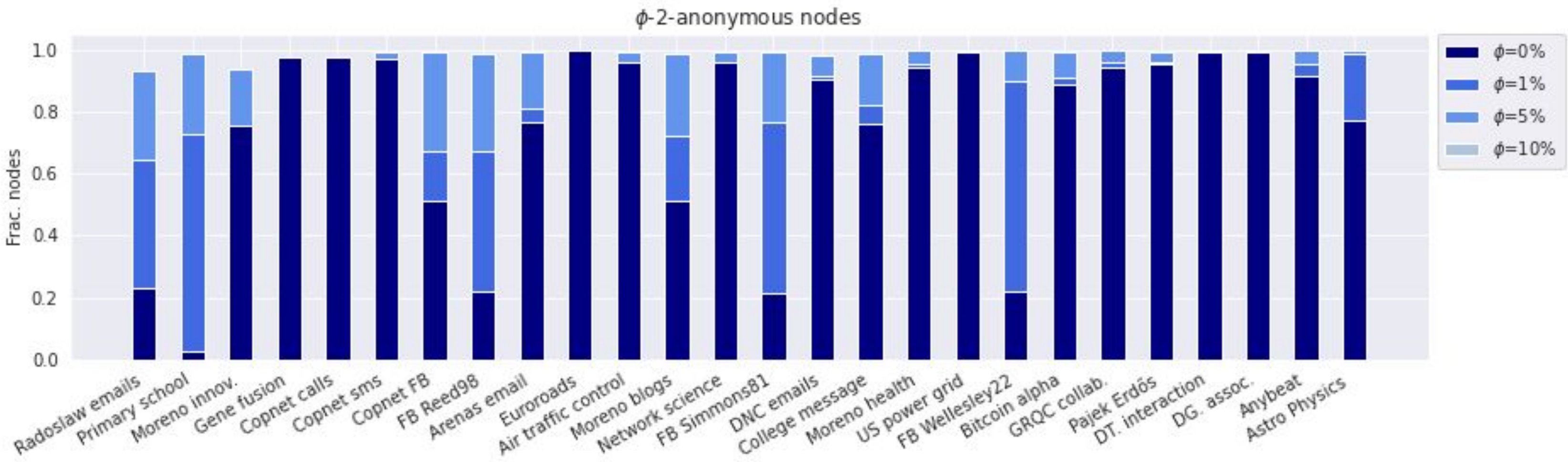
Fuzzy k-anonymity in networks



No uncertainty: **>2%** anonymous for all networks

Φ = 1%: **>60%** anonymous

Fuzzy k-anonymity in networks

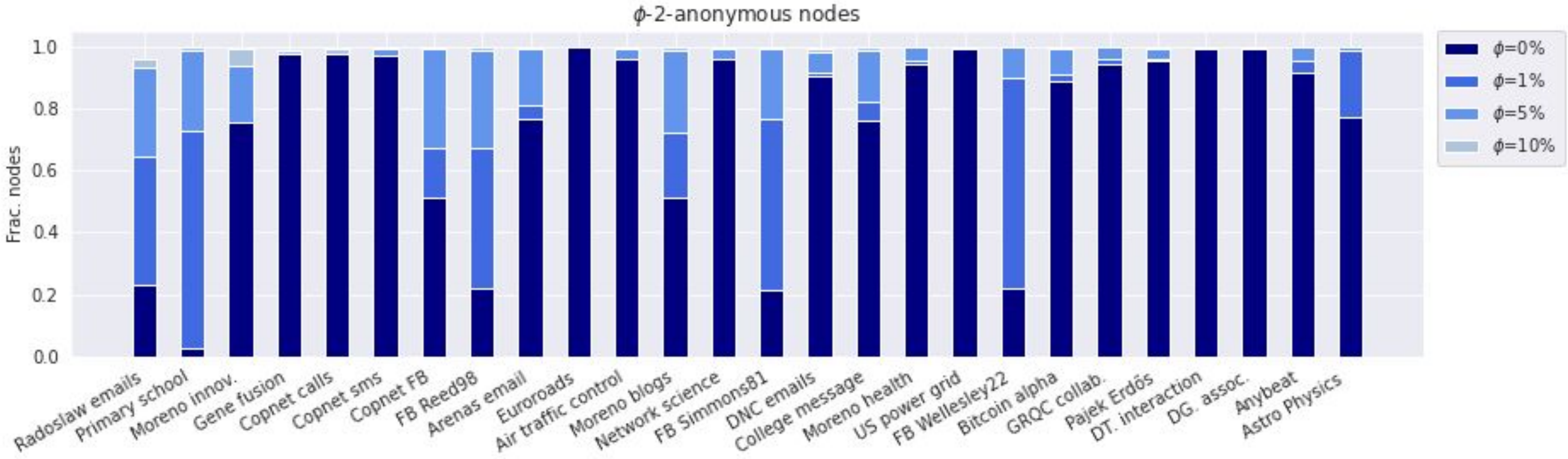


No uncertainty: **>2%** anonymous for all networks

Φ = 1%: **>60%** anonymous

Φ = 5%: **>90%** anonymous

Fuzzy k-anonymity in networks



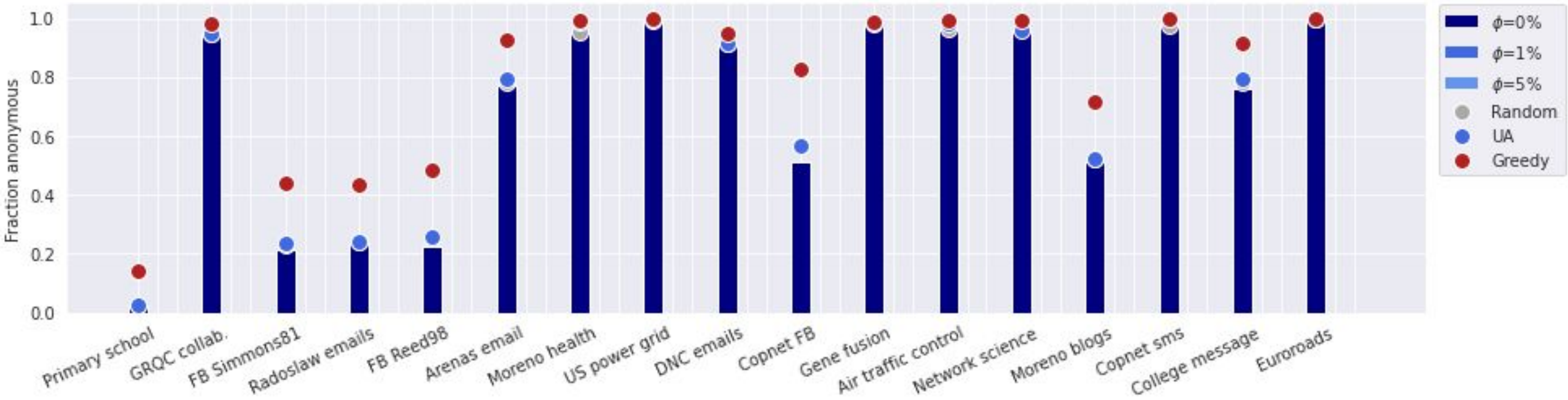
No uncertainty: **>2%** anonymous for all networks

Φ = 1%: **>60%** anonymous

Φ = 5%: **>90%** anonymous

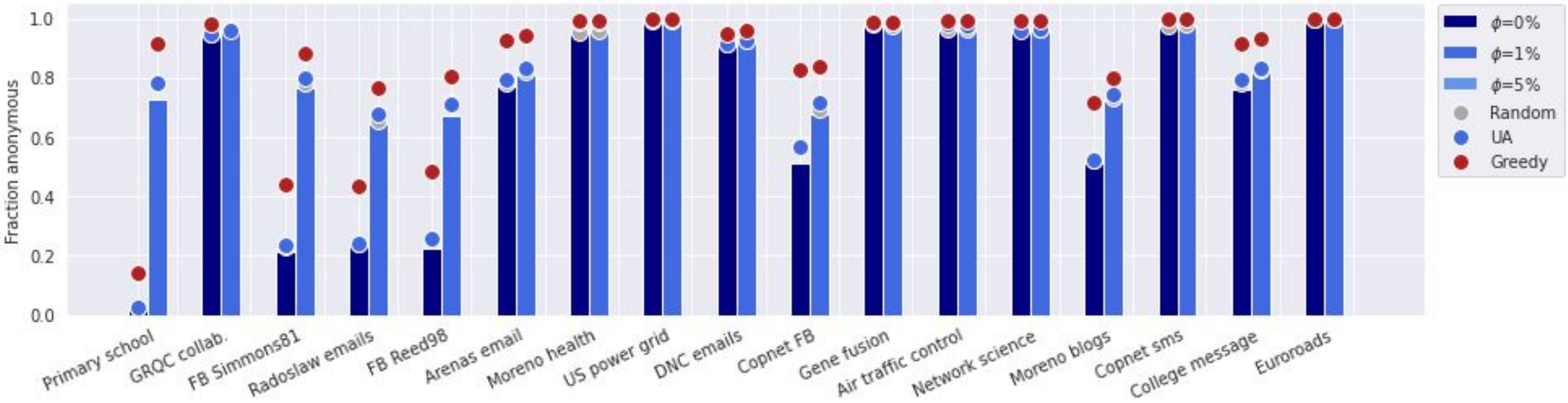
Φ =10%: **>95%** anonymous

Budgeted anonymization



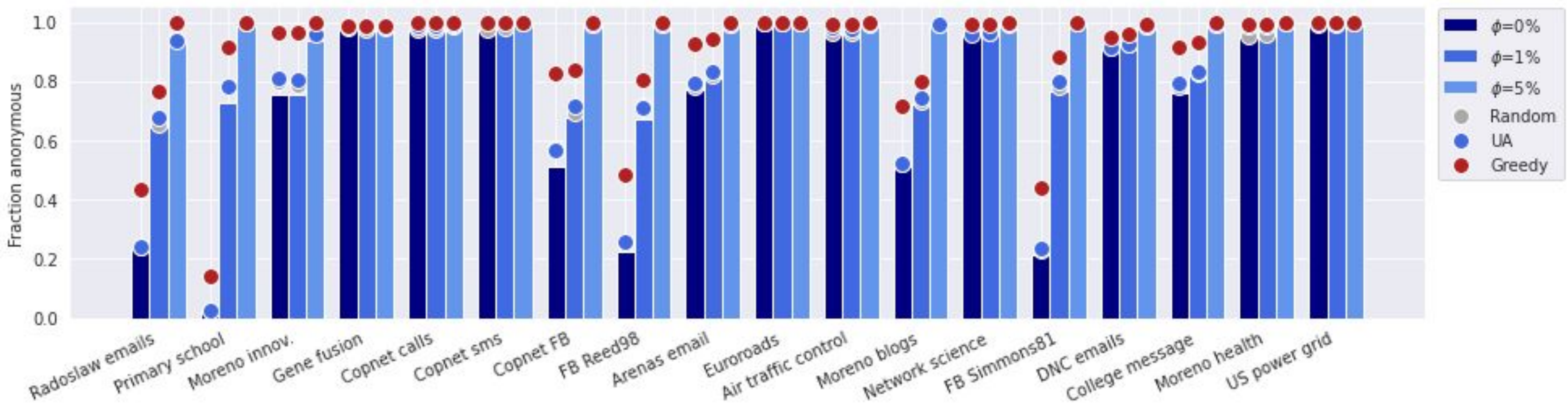
Anonymization: delete 5% of edges
No uncertainty: anonymization limited effect

Budgeted anonymization



Anonymization: delete 5% of edges
No uncertainty: anonymization limited effect
Φ = 1%: **>60%** anonymous

Budgeted anonymization



Anonymization: delete 5% of edges
No uncertainty: anonymization limited effect

Φ = 1%: **>60%** anonymous

Φ = 5%: **most networks fully anonymous**

Conclusion

- Measure anonymity with *k*-anonymity
- Attacker knowledge = anonymity measure
- Imprecise attacker knowledge
  $\longrightarrow$ large increase in anonymity
- Φ = 5% + anonymization
  $\longrightarrow$ Many networks fully anonymous

Future work

- Discussion: *which level of uncertainty is likely*?
  - Guidelines ϕ values
- Fuzzyness for other measures
- Include more network properties
  - Timestamps, node / edge labels

Rachel de Jong

Email

r.g.de.jong@liacs.leidenuniv.nl

linkedin.com/in/rachelgdejong/

ANONET package

github.com/RacheldeJong/ANONET

[1] **The effect of distant connections on node anonymity in complex networks.** Scientific Reports 14(1), 1156 (2024)
[2] **Algorithms for efficiently computing structural anonymity in complex networks.** ACM Journal of Experimental Algorithmics 28 (2023)
[3] **A systematic comparison of measures for k-anonymity in networks** (2024) arXiv:2407.02290
[4] **The anonymization problem in social networks**, preprint arXiv:2409.16163 (2024).
Authors: R.G. de Jong, M. P. J. van der Loo, F. W. Takes

Related research

How to **measure** anonymity?*

- **Algorithms for efficiently computing structural anonymity in complex networks**. *ACM Journal of Experimental Algorithmics*, *28*, 1-22. (2023).
- **The effect of distant connections on node anonymity in complex networks.** *Scientific Reports*, *14*(1), 1156. (2024)
- **A systematic comparison of measures for k-anonymity in networks.** *arXiv preprint arXiv:2407.02290*. (2024)
- Paper on fuzzy k-anonymity in progress…

How to **anonymize** a network?

The anonymization problem:

- de Jong, R. G., van der Loo, M. P., & Takes, F. W. **The anonymization problem in social networks**. *arXiv preprint arXiv:2409.16163*. (2024)

Genetic algorithm for anonymization:

- Bonello, S., de Jong, R. G., Bäck, T. H., & Takes, F. W. **Utility-aware Social Network Anonymization using Genetic Algorithms**. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion* (pp. 775-778). (2025)

Simulated Annealing for network anonymization:

- E.D. Arsene, R.G. de Jong, F.W. Takes, and A.L.D. Latour, **A Simulated Annealing Approach to Social Network Anonymization**. Accepted at Complex Networks (2025).

*Authors: Rachel G. de Jong, Mark P.J. van der Loo, Frank W. Takes